

Multi-model deep learning approach for segmentation of teeth and periapical lesions on pantomographs



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Introduction. The fields of medicine and dentistry are beginning to integrate artificial intelligence (AI) in diagnostics. This may reduce subjectivity and improve the accuracy of diagnoses and treatment planning. Current evidence on pathosis detection on pantomographs (PGs) indicates the presence or absence of disease in the entire radiographic image, with little evidence of the relation of periapical pathosis to the causative tooth.

Objective. To develop a deep learning (DL) AI model for the segmentation of periapical pathosis and its relation to teeth on PGs.

Method. 250 PGs were manually annotated by subject experts to lay down the ground truth for training AI algorithms on the segmentation of periapical pathosis. Two approaches were used for lesion detection: Multi-models 1 and 2, using U-net and Mask RCNN algorithms, respectively. The resulting segmented lesions generated on the testing data set were superimposed with results of teeth segmentation and numbering algorithms trained separately to relate lesions to causative teeth. Hence, both multi-model approaches related periapical pathosis to the causative teeth on PGs.

Results. The performance metrics of lesion segmentation carried out by U-net are as follows: Accuracy = 98.1%, precision = 84.5%, re-call = 80.3%, F-1 score = 82.2%, dice index = 75.2%, and Intersection over Union = 67.6%. Mask RCNN carried out lesion segmentation with an accuracy of 46.7%, precision of 80.6%, recall of 55%, and F-1 score of 63.1%.

Conclusion. In this study, the multi-model approach successfully related periapical pathosis to the causative tooth on PGs. However, U-net outperformed Mask RCNN in the tasks performed, suggesting that U-net will remain the standard for medical image segmentation tasks. Further training of the models on other findings and an increased number of images will lead to the automation of the detection of common radiographic findings in the dental diagnostic workflow. (Oral Surg Oral Med Oral Pathol Oral Radiol 2024;138:196–204)

Pantomographs (PGs) are the most common imaging modality used for the analysis of dentition and related anatomy for a complete dental evaluation.¹ They provide a panoramic view of teeth, maxillae, mandible, maxillary sinuses, temporomandibular joints, etc.¹ Some radiographic findings are incidental on PGs, especially if the patient is asymptomatic, thus indicating when further area-specific evaluation is needed.² An example is a periapical pathosis, caused by apical periodontitis, which presents as a radiolucent area around the apices of teeth.³ The detection of periapical pathosis on PG is 77% sensitive and 95% specific; therefore, when detected, a periapical radiograph can be obtained for a more detailed examination.⁴ Hence, PGs have good overall diagnostic discriminatory ability and is employed in daily clinical practice as a screening tool.^{5,6}

Computer vision (CV) is a field of artificial intelligence (AI) that allows computers to derive meaningful information from images, with its consequent application to solve problems.^{7,8} Convolutional neural

networks (CNNs) are the algorithms of choice when applying AI techniques to images.^{9,10} Convolutional neural networks are robust at a variety of CV tasks, including object detection, segmentation, and classification (Figure 1).^{10,11} These tasks can be applied to radiographs for the detection of normal anatomy as well as pathosis for developing an automated clinical decision support system. Therefore, the main objective of our research is to train AI to identify and list all possible findings on PGs, including normal and pathologic conditions. The first step toward this goal was to train an AI model to perform teeth segmentation and numbering tasks, details of which can be found in the cited text.¹² In this paper, we describe the second step towards automation, which involves the detection of periapical pathosis and its relation to teeth on PGs. Once adequately trained, this tool will screen for periapical pathosis and alert clinicians to investigate possibly diseased teeth with further area-specific radiographs as well as clinical correlations and/or tests. Previous literature has described the use of AI for detecting periapical pathosis on radiographs, but they only detect the presence or absence of disease

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Statement of Clinical Relevance

The application of artificial intelligence for the detection of periapical disease on pantomographs with the causative tooth identified directs a clinician toward further area-specific examination to confirm the presence and extent of disease.

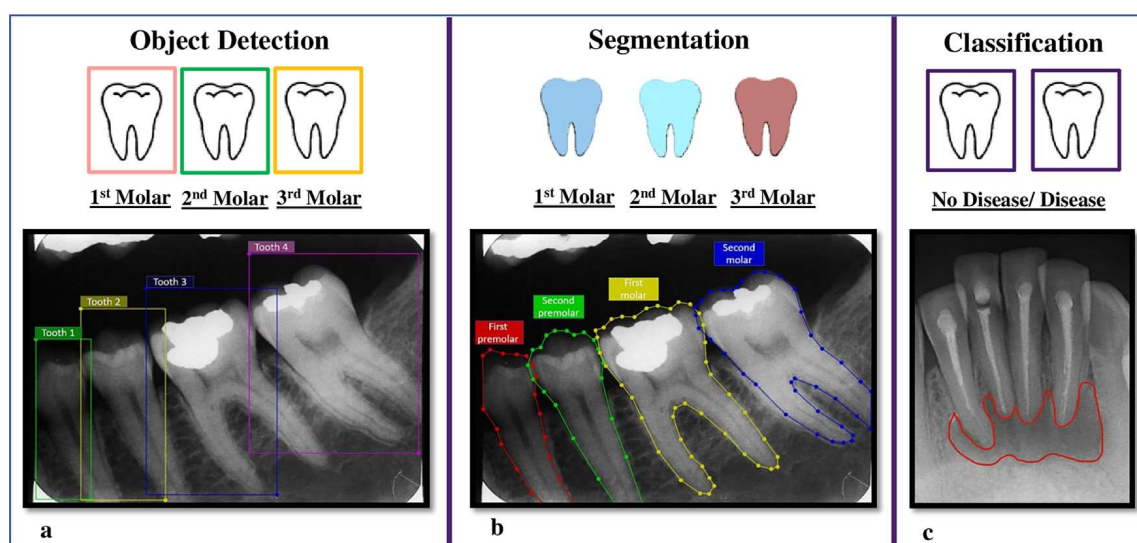


Fig. 1. (A) Object detection: teeth are localized using bounding boxes. (B) Semantic segmentation: teeth are separated from the background using fluid outlines. (C) Classification; diseased and healthy teeth are classified separately.

in the entire radiograph without relating it to relevant anatomic structures.^{2,13}

This paper explains an extension of our previous work,¹² and two approaches were used to detect and relate periapical pathosis to specific teeth on PGs. In the future, the expansion of this work will lead to the development of AI-based diagnostic software that will improve the overall efficiency of identification of common pathoses in the dental diagnostic workflow.

METHOD

This study was conducted at Aga Khan University Hospital, Karachi, Pakistan, from May 2022 to August 2022. Pantomographs were retrieved from the radiographic database of dental clinics after obtaining an exemption from the ethics review committee (2022-8083-23090) of the institution. All PGs made during the mentioned period were reviewed, and only those fulfilling the inclusion criteria were used in the study. Pantomographs from an open-sourced data set, TUFTS Dental Database (TDD), were also used to train and test the algorithms.¹⁴ The sample size for training the algorithm was determined arbitrarily, and training was terminated when the algorithms reached near saturation. Pantomographs were selected via a non-probability purposive sampling technique according to the set inclusion and exclusion criteria as follows.

Inclusion criteria

- Good quality pantomographs
- Teeth with closed apices
- Fully erupted permanent dentition
- Periapical pathosis associated with one or more teeth

Exclusion criteria

Pantomographs with the following.

- Artifacts
- Impacted teeth
- Supernumerary teeth
- Deciduous teeth
- Orthodontic appliances, titanium plates, and wires
- Generalized root resorption post orthodontic treatment

Pantomographs obtained from the locally sourced data set were scanned by Orthophos XG 3D (Dentsply Sirona GmbH) operated at 60 to 90 kV and 3 to 16 mA, with image dimensions of 2,440 × 1,292 pixels, captured in gray level. The PGs were downloaded from SIDEXIS XG 2.63 (Dentsply Sirona GmbH) software in PNG format from the image database. The PGs were anonymized, and patient metadata was unavailable. All PGs were kept in a database, which was password-protected and only accessible to the authors of this study. Pantomographs from TDD were also used; the owners of this data set provided masks to be employed in training; however, for the purpose of this study, we carried out our own annotations to maintain accuracy and uniformity in annotations.¹⁴

Dataset annotation

The images comprising the dataset were in two formats: PNG for locally sourced data set and JPEG for TDD. A segmentation map for every tooth was created manually via polygons using the Labelme software, which was downloaded from GitHub.¹⁵ Each of the 250 radiographs was labeled, and annotations were generated in JSON format to be used for training.

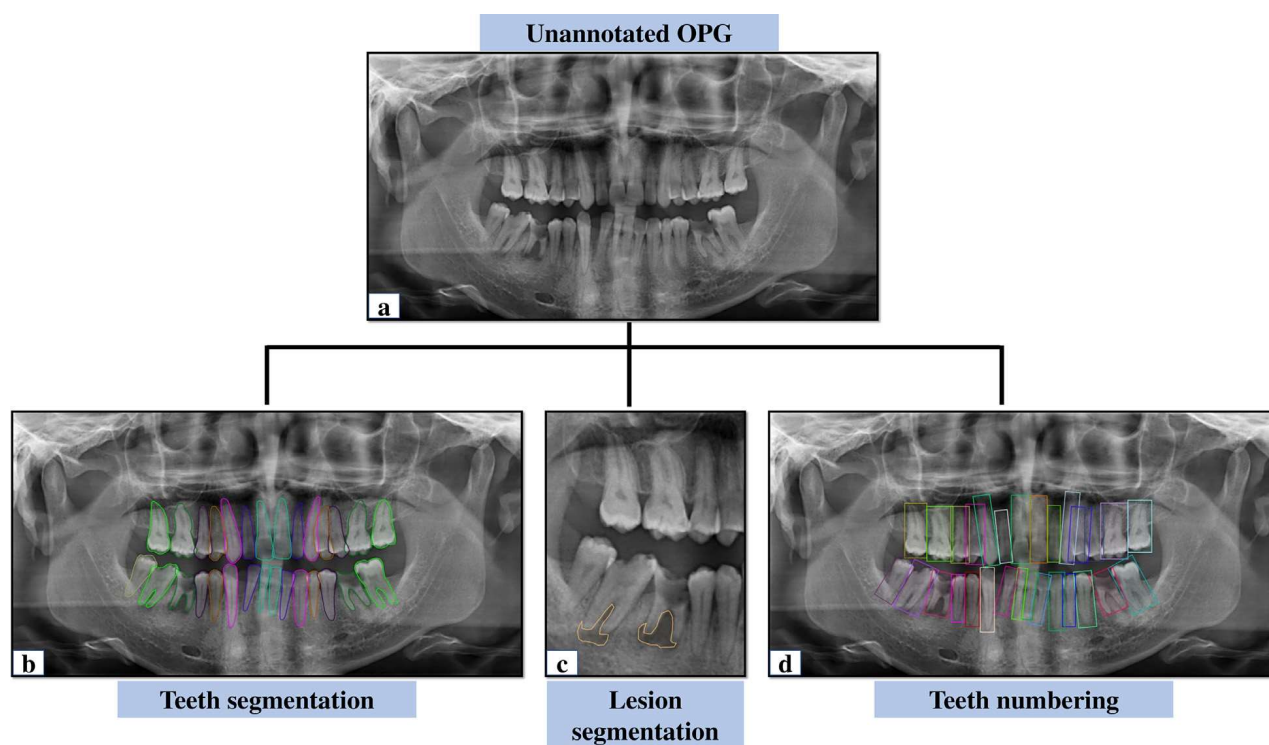


Fig. 2. (A) Unannotated, raw Panoramograph (PG) to be annotated. (B) PG with teeth annotated using polygons to separate each pixel of the tooth from the background. (C) Periapical lesions annotated using polygons after being zoomed in for better visualization. (D) PG with teeth annotated using bounding boxes according to the Fédération Dentaire Internationale numbering system.

Every PG was annotated for three tasks, as shown in Figure 2, namely, individual teeth segmentation according to the class (molar, premolar, etc.) and teeth numbers using bounding boxes according to the Fédération Dentaire Internationale numbering system (18, 17, 16, etc.). These formed the ground truth for training algorithms for teeth segmentation and teeth numbering tasks. The same cohort of 250 PGs was also annotated for periapical pathosis separately using the same software. The annotations were carried out, denoting lesions at apices of teeth by zooming in the PGs by 200% for better visualization (Figure 2). This separate annotated data set formed the ground truth for training algorithms for the lesion segmentation task.

All annotations were carried out by two investigators of this study: a postgraduate student and a specialist endodontist. The annotations by the first investigator were checked by the second investigator, and vice versa; any conflicts in the annotations were resolved via discussion. All investigators were calibrated on the use of the software before beginning the annotation process. The performance of all algorithms was evaluated using the testing data set after training was complete. This study was performed in accordance with the AI in dental research checklist by Schwendicke et al.; the following narrative explains two techniques used

by authors to carry out lesion segmentation and consequently relate it to the causative tooth on PG.¹⁵ The following narrative describes the use of two multi-models for relating periapical pathosis to causative teeth.

Multi-model 1. Three CNNs were used in this approach for the segmentation and numbering of all teeth as well as the segmentation of periapical pathosis on PGs. Two CNNs (U-net and Faster RCNN) were previously trained for teeth segmentation and numbering¹² (Figure 3); additionally, another U-net was trained on periapical pathosis detection.¹⁶ The three algorithms were run in parallel (Figure 3), and the results of each were superimposed via Python code written specifically for the purpose of superimposition. The details of the 3 algorithms are as follows:

1. U-net A: Teeth segmentation algorithm previously trained (further trained on 250 PGs)
2. Faster RCNN: Teeth numbering algorithm previously trained (further trained on 250 PGs)
3. U-net B: Lesion segmentation algorithm

The basic U-net algorithm employed, an in-depth explanation of the workings of the algorithms, is beyond the scope of this paper and can be found elsewhere.¹⁷

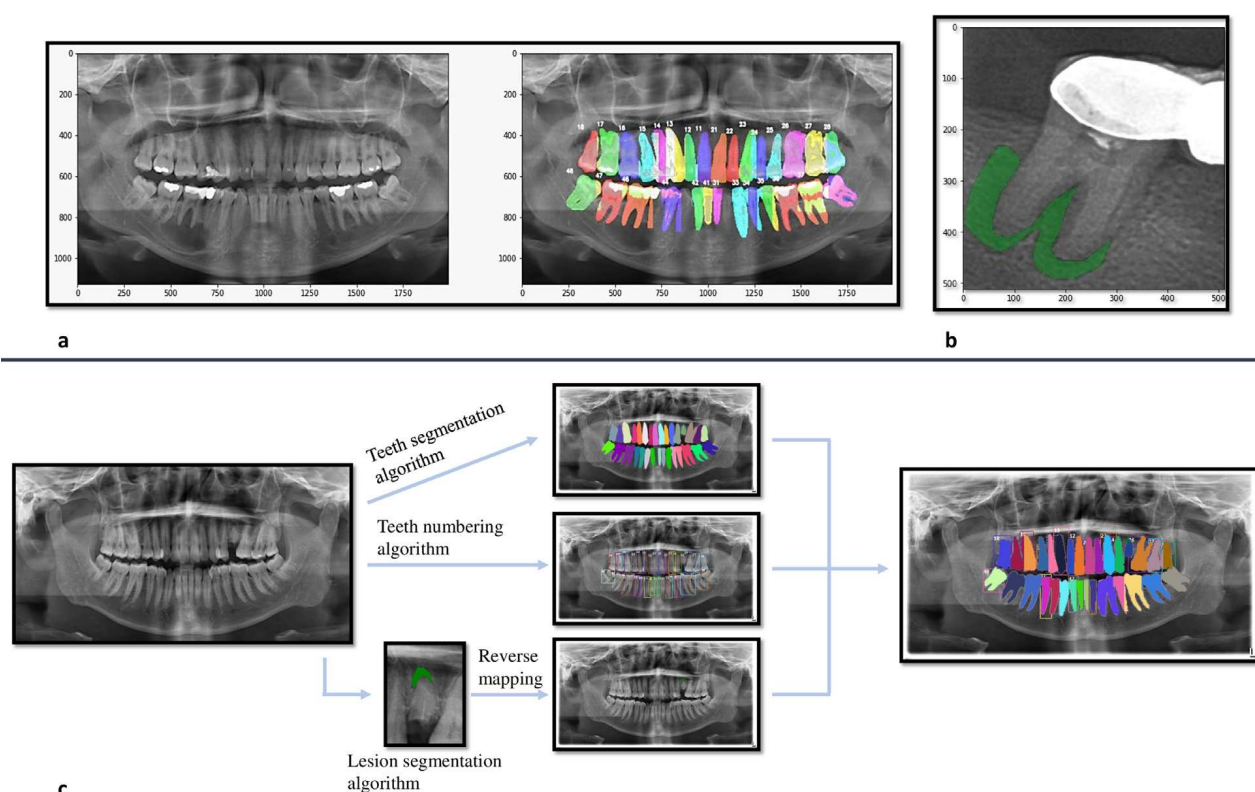


Fig. 3. (A) Teeth segmentation and numbering results of U-net A and Faster RCNN combined. (B) Lesion segmentation results of U-net (C) Workflow of multi-model 1 with results of all 3 algorithms combined.

Before training, each radiograph was resized from the original dimensions of $2,440 \times 1,292$ pixels to 512×512 pixels to account for the limited computational resources. The training was carried out on GoogleColabPro (Google, LLC), with random initialization of parameters.¹⁸

The previously trained algorithms, U-net A and Faster RCNN, were further trained and tested on 250 annotated images with the method described in the previous paper (Figure 3).¹² Additionally, the U-net B algorithm was trained and tested on the same set of 250 PGs using similar parameters as the previous algorithms for lesion detection tasks.¹² However, complete PGs were not used for training this algorithm; instead, the bounding boxes generated from Faster RCNN formed the training data set for this algorithm. To ensure that lesions were captured adequately within the outlines of these boxes, the generated bounding boxes were enlarged by 100 pixels. These enlarged images formed individual tooth “crops” that were used to train U-net B. A total of 1,354 crops (from 250 PGs) were used to train U-net B; the images were augmented using techniques stated in the previous study.¹² Out of 1,354 images, 1,273 were used for training the algorithm, 13 for validation, and 68 were used for testing the performance of the algorithm; this split was carried out randomly by the algorithm. It is important to note

that out of the 1,354 crops, 747 crops contained periapical lesions, out of which 230 contained lesions associated with endodontically treated teeth. This indicates that the disease prevalence in our data set was almost 50%.

The algorithm was trained using an Adam optimizer with a binary cross-entropy loss function. It was trained for 50 epochs, with each epoch taking 160 iterations. The training process took 2.2 hours, with an average of 158.4 seconds per epoch with a batch size of 8.

Because U-net B was trained on crops, these had to be “reverse mapped” onto whole OPGs to be superimposed on the visual results of U-net A and Faster RCNN, as shown in Figure 3 (see Table I for performance metrics).

RESULTS (MULTI-MODEL 1)

The algorithms were successful in achieving the desired tasks at reasonable performance metrics. The visual results are shown in Figure 3. Various performance metrics were used to gauge the performance of U-net A, U-net B, and Faster RCNN algorithms.

The performance of the existent teeth segmentation, U-net A, was further improved as indicated by the following performance metrics, including accuracy = 98.1%, precision = 91.8%, recall = 93.3%, F-1 score = 92.5%, dice

Table 1. Performance metrics of teeth segmentation and numbering tasks

<i>Teeth segmentation and numbering tasks</i>			
<i>Performance metrics</i>	<i>U-net A</i>	<i>Faster RCNN</i>	<i>Mask RCNN A</i>
Dice index	90.1%	—	—
IoU	82.1%	—	—
Accuracy	98.1%	93.9%	54.8%
F-1 score	92.5%	—	70.6%
Re-call	93.3%	—	54.8%
Precision	91.8%	—	100%
Mean overlap of bounding box	—	16.346	—

index = 90.1%, and Intersection over Union (IoU) = 82.1% for U-net A. The performance metrics of the Faster RCNN algorithm were determined using overlap accuracy = 16.3 bounding boxes and classifier accuracy of labels = 93.9%. This is relative to the previous performance metrics when it was trained and tested on only 50 OPGs.¹²

The performance metrics of the lesion segmentation algorithm carried out by the current model, U-net B, are as follows: Accuracy = 98.1%, precision = 84.5%, recall = 80.3%, F-1 score = 82.2%, dice index = 75.2%, and IoU = 67.6%.

(An explanation of the metrics can be found in the [Appendix](#).)

Multi-model 2

Two Mask RCNN algorithms formed multi-model 2. These were trained on a computer equipped with Nvidia GTX-1060 (Nvidia, Corp.), a CUDA-based GPU, on Tensorflow 2.0 (Google Brain Team), paired with Intel's 8th Generation Core i7 CPU and 16 GB DDR4 RAM (Intel, Corp.). The backbone for Mask R-CNN used in this model was ResNet50.¹⁹ The initial weights for the model were loaded from the COCO data set.²⁰ The training process was run continuously until the marginal improvement between epochs was below the 1% threshold. This multi-model comprises 2 algorithms trained on 250 PGs, randomly split into training, validation, and testing data sets by the algorithms, performing the following tasks:

1. Mask RCNN A: Mask RCNN for teeth segmentation and numbering
2. Mask RCNN B: Mask RCNN for lesion segmentation

These algorithms were trained using the stochastic gradient descent optimizer with a sigmoid loss function.

The teeth segmentation and numbering Mask RCNN algorithm (Mask RCNN A) was run for 310 epochs,

with each epoch taking 100 iterations. In addition, the data were augmented for this model to include variety as well as expand the number of examples.

The data was augmented using the following techniques.

1. Cropping and padding each side of the lesion polygons by 25%
2. Blend in the Alpha channel and remove about 50% of the color
3. Flip random 50% of the images horizontally and random 50% vertically
4. Scale images from 75% to 125% of their original size
5. Rotate images between -30° and 30°
6. Sheer images by -15° to 15°

The training process took 12 hours, averaging 135.4 seconds per epoch. The batch size for training the algorithm was two, and the images were scaled to a maximum width and height of 832 pixels. Each tooth was identified by a label, according to the Fédération Dentaire Internationale numbering system. Therefore, 32 classes were used for training. All boxes generated from this algorithm were saved separately, forming the "crops" to train the other algorithm.

The crops generated from Mask RCNN A (153 in total) formed the training data set for lesion detection Mask RCNN algorithm (Mask RCNN B). The Mask RCNN B algorithm was trained for 500 epochs, each running for 100 iterations. This model took 11 hours of training, averaging 63.8 seconds per epoch. The batch size for this model was one, whereas the other parameters remain similar to that of Mask RCNN A.

Once trained, both models were sequentially run to predict teeth segmentation and numbering, as well as lesions on the same test images. Both outputs are superimposed for combined feedback, such that the output from Mask RCNN A is overlapped on Mask RCNN B output with 50% transparency ([Figure 4](#)) (see [Table 1](#) for performance metrics).

RESULTS (MULTI-MODEL 2)

Aside from the overall evaluation, we used the Tensorboard utility, provided by the TensorFlow toolkit, to measure how the models progressed over time.

The performance metrics of Mask RCNN A included an accuracy of 54.8%, precision of 100%, recall of 54.8%, and F-1 score of 70.6%. Similarly, the performance metrics of Mask RCNN B included an accuracy of 46.7%, precision of 80.6%, recall of 55%, and F-1 score of 63.1%.

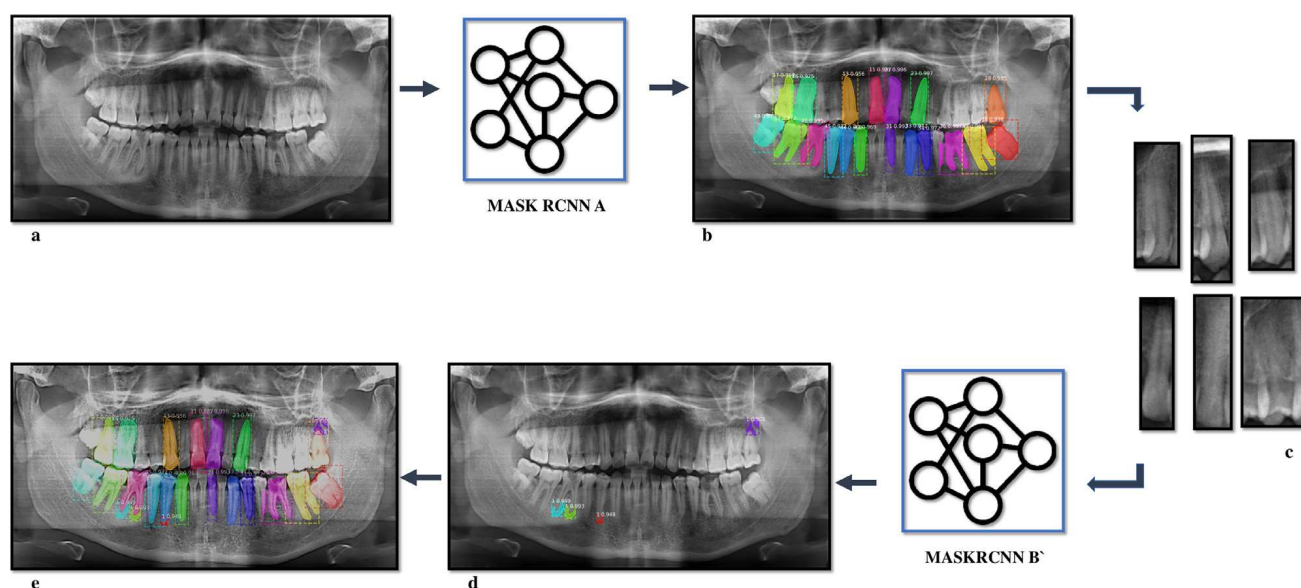


Fig. 4. (A) An unseen image ran through multi-model 2. (B) Teeth segmentation and numbering results of Mask RCNN A. (C) Teeth crops generated from Mask RCNN A formed the data set for Mask RCNN B. (D) Lesion segmentation results which crops reverse mapped back to form a whole Pantomograph. (E) Composite result of Mask RCNN A and Mask RCNN B.

A comparison of results based on tasks

Tables I and II compare the performance relative to the specific tasks performed. The combination of U-net A and Faster RCNN carries out teeth segmentation and numbering tasks with higher performance compared with Mask RCNN A. U-net B algorithm segments lesions with better performance than Mask RCNN B. In our study, both U-nets outperformed the other algorithms in their respective tasks.

A direct comparison of both U-nets shows the higher performance of U-net A (teeth segmentation) compared with U-net B (lesion segmentation). This is assumed to be due to the increased number of examples of “teeth” on PG compared with the fewer number of “lesions” used to train the algorithms.

Both lesion segmentation algorithms U-net B and Mask RCNN B were trained on the crops generated from the preceding teeth numbering algorithms. Here, the authors would like to emphasize the difference in the number of crops used for training each algorithm.

Table II. Performance metrics of lesion segmentation task

<i>Lesion segmentation task</i>		
<i>Performance metrics</i>	<i>U-net B</i>	<i>Mask RCNN B</i>
Dice index	75.2%	—
IoU	67.6%	—
Accuracy	98.1%	46.7%
F-1 score	82.2%	63.1%
Re-call	80.3%	55%
Precision	84.5%	80.6%

Whereas U-net B was trained on 1,354 crops, Mask RCNN B was trained on only 153 crops. This vast difference was due to the different accuracies of the teeth numbering algorithms generating the crops for each algorithm. Therefore, extensive augmentation of data for Mask RCNN to account for this difference but the results of this algorithm are still relatively poor.

DISCUSSION

With the advancement of AI techniques in biomedical imaging, there has been an upsurge in its application in radiographic diagnosis. Among these, the identification of normal anatomy and the detection of pathosis have been well investigated in the literature.^{11,13} However, the relation of pathosis with associated anatomy has not been explored thoroughly. Similarly, this gap has also been observed in dental radiomics, where the association of periapical disease with teeth is not well established.^{2,13,21-23} The localization of periapical pathosis to causative tooth is a critical element of dental diagnosis. Real-life applicability can be achieved if the origin of pathosis and associated anatomy are related. The detection of teeth alone or the presence/absence of pathosis in an entire image carries no benefit for clinicians. From a clinical standpoint, the relation of pathosis to teeth outweighs the high-performance metrics of the algorithm in terms of providing better patient health care.

A recent systematic review by Sadr et al. found that DL models had acceptable accuracy for detecting lesions on periapical radiographs, PGs, and cone beam

computed tomography (CBCT) images.¹³ However, the authors of the included studies did not analyze whether the disease (i.e., periapical pathosis was related to causative anatomy).¹³ In a study by Yuksel et al., they designed a DL model that performs teeth numbering tasks and also recommends treatment needs of individual teeth, thus overcoming some of the shortcomings in current literature.²⁴ The main objective of this study was to develop a DL algorithm for periapical disease detection along with detection and relation teeth. Because the task was multifold, the authors designed a robust method including multiple algorithms. The first goal of this study was to train an AI model that can perform teeth segmentation and numbering on PGs with high accuracy, which the authors have achieved.¹² In continuation with the previous work, another finding (i.e., periapical pathosis) was added to the model, which resulted in the segmentation of periapical pathosis and improved performance of the teeth segmentation and numbering tasks. In the execution of this task, a heterogeneous data set of PGs was used for training (i.e., images with missing teeth, restorative and prosthetic work, carious lesions, etc.). To incorporate variation in our data set and to improve the generalizability of the models, images from TDD were also included.¹⁴

In this study, two models were trained for lesion detection; both are examples of “ensemble learning.”²⁵ This is the combination of multiple algorithms, usually carried out to improve the accuracy of results of the individual algorithms. It includes techniques such as bagging, boosting, stacking, and blending to enhance accuracy.²⁵ However, the integration of visual results also comes under the umbrella of an “ensemble,” and results from each of the algorithms are stacked for a final result.²⁶ Alternatively, the models in our experiment can also be classified as “multi-model” deep neural networks because multiple algorithms are trained independently on the same data set to learn different representations and generate complementary results.²⁷

For multi-model 1, we aimed to improve on our existing algorithms as well as add another algorithm, U-net B, for lesion detection and achieved an accuracy of 98.1%. This model comprises three algorithms running in parallel, which were consequently combined to relate periapical pathosis to the causative tooth. To achieve optimal performance, each algorithm was trained to execute only one task. Because U-net B was trained on crops, these had to be reverse-mapped back onto whole OPGs to generate an image compatible for superimposition with the other 2 algorithms. This multi-model used U-net algorithms which are known to specialize in image segmentation tasks and with relatively smaller data sets for training, which is adequately reflected in the results.

In an attempt to improve computational efficiency, Mask RCNN was used to reduce the number of algorithms required to achieve tasks. Mask RCNN can perform all 3 tasks (Figure 1) concurrently, as opposed to U-net (teeth segmentation) and Faster RCNN (teeth numbering). Multi-model 2 comprised two similar algorithms running in series, one performing teeth segmentation and numbering (Mask RCNN A) and the other segmenting lesions (Mask RCNN B), with consequent superimposition of results. The outputs generated by Mask RCNN A formed the training data set for Mask RCNN B, and the output of Mask RCNN B was reverse-mapped onto whole OPGs for a consolidated result, similar to multi-model 1. The accuracy of lesion segmentation of this model, however, was only 46.7% even having used a more versatile algorithm. He et al. compared Mask R-CNN with other state-of-the-art object detection algorithms on the COCO data set and concluded that Mask R-CNN outperformed other algorithms in image segmentation.²⁸ Therefore, the authors of this study used the weights from a Mask RCNN algorithm pre-trained on the COCO data set; however, the results are contrary.^{19,20} This could be attributed to the fact that He et al. did not carry out these tasks on radiographs, as was the case in this study.¹⁹ In this model, the data set was also augmented in an attempt to improve performance, but the results remained unsatisfactory. Therefore, in our experiment, the U-net approach has been successful in lesion segmentation thus far.

Both lesion segmentation algorithms of multi-models 1 and 2 were trained on crops generated from their respective teeth segmentation and numbering algorithms. This was done to eliminate confounding anatomic landmarks such as mental foramen from data that can be mistaken for periapical pathosis. Using crops also increased the number of images comprising the training data set. The use of the reverse-mapping technique led to the generation of composite results of both multi-models, 1 and 2.

The high performance of U-net has been consistent, as evidenced by Setzer et al. and Zheng et al., who achieved a dice index of 67% and 74% on lesion detection on CBCT images, respectively.^{29,30} This is in comparison with the dice index score of 75.2% achieved by U-net B trained in this study. Bayrakdar et al. also used U-net to detect periapical pathosis on PG with an IoU of 70% using 470 images, compared with 67.6% in this study.²²

A recent study by Fatima et al. successfully trained a Mask RCNN for lesion detection with an accuracy of 94%; however, they used periapical images in their data set.³¹ In comparison, Mask RCNN B in this study achieved an accuracy of 46.7% on OPG; the low accuracy may be attributed to the lower resolution of PG relative to periapical radiographs, which can make lesion segmentation a difficult task to execute.

The strengths of this study include the use of a multi-model approach to successfully relate periapical pathosis to the causative teeth. The use of a locally sourced data set, as well as TDD, enhanced the diversity of the data in this study.¹³ As mentioned in the recent systematic review by Sadr et al., lesions are difficult to detect, especially on PG, due to their smaller size and poorly defined borders. Therefore, in an attempt to account for this deficiency, the PGs were zoomed to 200% during annotation for better visualization of lesions. Crops were used rather than whole PGs to increase the diagnostic yield of both lesion detection algorithms (U-net B and Mask RCNN B).

One of the limitations of our study, however, is the lack of general applicability of results due to the strict inclusion criteria where all PGs included have periapical pathosis. This led to the incorporation of selection bias and the number of crops with and without lesions being at almost 1:1. As mentioned previously; this indicates that the disease prevalence in our data set is higher than that found in the general population. Therefore, further training on a data set with disease prevalence that is reflective of the population is needed. The authors emphasize that an PG is not diagnostic for periapical pathosis, and the clinical presentation of these lesions could not be accounted for due to the nature of this study. However, the detection of a periapical lesion directs the clinician to further investigate the region using area-specific radiographs and clinical testing. Furthermore, our results cannot be generalized to entire populations, for example, pediatric patients, because these PGs were excluded. Our models are at stage 2 of AI development, according to Zhang et al., and all the aforementioned limitations are usually addressed at stages 3 and 4, which include model deployment that the authors are working toward.³² In the future, authors aim to further train the algorithms on current diagnostic tasks and also to train AI for the detection of additional pathoses on PG with the ultimate goal of automation. At that stage, a head-to-head comparison of AI methods with humans can also be performed to determine the actual performance of these models in real-life settings. Moreover, there is a lack of data on how AI may alter the diagnostic process and decision-making or how it may impact cost-effectiveness.³³ Once AI development reaches stage 4, this crucial information will be determined, and a true cost-benefit analysis of AI in health care will be carried out effectively.

CONCLUSIONS

Both models were able to perform teeth segmentation and numbering tasks on PG as well as relate the pathosis to causative teeth. However, U-net was

consistently better at the tasks it was being trained for, with 98.1% accuracy. Even though Mask RCNN is a more versatile and complex algorithm, it failed to deliver the same robustness and accuracy (46.7%) compared with U-net. Within the limitations of this study, the authors concluded that U-net may remain the algorithm of choice for medical image segmentation until a more robust algorithm proves otherwise. The code for both algorithms can be found using the following link: <https://github.com/Niihhaa/lesion-detection-both-algorithms-.git>.

DISCLOSURE

None.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.oooo.2023.11.006](https://doi.org/10.1016/j.oooo.2023.11.006).

REFERENCES

1. Silva G, Oliveira L, Pithon M. Automatic segmenting teeth in X-ray images: trends, a novel data set, benchmarking and future perspectives. *Expert Syst Appl*. 2018;107:15-31.
2. Ekert T, Krois J, Meinhold L, et al. Deep Learning for the radiographic detection of apical lesions. *J Endod*. 2019;45: 917-922.e5.
3. Segura-Egea JJ, Martín-González J, Castellanos-Cosano L. Endodontic medicine: connections between apical periodontitis and systemic diseases. *Int Endod J*. 2015;48:933-951.
4. Arslan ZB, Demir H, Berker Yıldız D, Yaşar F. Diagnostic accuracy of panoramic radiography and ultrasonography in detecting periapical lesions using periapical radiography as a gold standard. *Dentomaxillofac Radiol*. 2020;49:20190290.
5. Nardi C, Calistri L, Grazzini G, et al. Is panoramic radiography an accurate imaging technique for the detection of endodontically treated asymptomatic apical periodontitis? *J Endod*. 2018;44:1500-1508.
6. Maddalone M, Bonfanti E, Pellegatta A, Citterio CL, Baldoni M. Digital orthopantomography vs cone beam computed tomography-Part 1: detection of periapical lesions. *J Contemp Dent Pract*. 2019;20:593-597.
7. Xu Y, Wei H, Lin M, et al. Transformers in computational visual media: a survey. *Comput Vis Media*. 2022;8:33-62.
8. Umer F, Habib S. Critical analysis of Artificial Intelligence in endodontics: a scoping review. *J Endod*. 2022;48:152-160.
9. Lin T, Wang Y, Liu X, Qiu X. A survey of transformers. *AI Open*. 2023;3:111-132.
10. Adnan N, Umer F. *Understanding deep learning—challenges and prospects*. 2022;72(Suppl 1):S59-S63.
11. Umer F, Habib S, Adnan N. Application of deep learning in teeth identification tasks on panoramic radiographs. *Dentomaxillofac Radiol*. 2022;51:20210504.
12. Adnan N, Khalid WB, Umer F. An artificial intelligence model for teeth segmentation and numbering on orthopantomograms. *Int J Comput Dent*. 2023;26:301-309.
13. Sadr S, Mohammad-Rahimi H, Motamedian SR, et al. Deep Learning for detection of periapical radiolucent lesions: a

- systematic review and meta-analysis of diagnostic test accuracy. *J Endod.* 2023;49: 248-261.e3.
14. Panetta K, Rajendran R, Ramesh A, Rao S, Agaian S. Tufts Dental Database: a multimodal panoramic X-Ray dataset for benchmarking diagnostic systems. *IEEE J Biomed Health Inform.* 2022;26:1650-1659.
 15. Schwendicke F, Singh T, Lee JH, et al. Artificial Intelligence in dental research: checklist for authors, reviewers, readers. *J Dent.* 2021;107:103610.
 16. UNet Line By Line Explanation: Towards Data Science. Medium. Accessed July 15, 2021. <https://towardsdatascience.com/unet-line-by-line-explanation-9b191c76baf5>.
 17. Ronneberger O, Fischer P, Brox T. U-net: convolutional networks for biomedical image segmentation. *Int Conf on MIC-CAI*. 2015:234-241.
 18. Welcome To Colaboratory. Google Colaboratory. Accessed July 15, 2021. <https://colab.research.google.com/notebooks/intro.ipynb>.
 19. He K, Gkioxari G, Dollar P, Girshick R, Mask R-CNN. In: *Proceedings of the IEEE International Conference on Computer Vision*; 2017. p. 2961-2969.
 20. Lin T-Y, Maire M, Belongie S, eds. *Microsoft coco: common objects in context. Computer Vision—ECCV 2014: 13th European Conference*, Springer, Zurich, Switzerland; 2014. September 6-12, 2014, *Proceedings*, Part V 13.
 21. Leite AF, Gerven AV, Willems H, et al. Artificial Intelligence-driven novel tool for tooth detection and segmentation on panoramic radiographs. *Clin Oral Investig.* 2021;25:2257-2267.
 22. Bayrakdar IS, Orhan K, Çelik Ö, et al. A U-Net approach to apical lesion segmentation on panoramic radiographs. *Biomed Res Int.* 2022;2022:7035367.
 23. Ngoc V, Viet DH, Anh LK, et al. Periapical lesion diagnosis support system based on X-ray images using machine learning technique. *World J Dent.* 2021;12:190.
 24. Yüksel AE, Gültekin S, Simsar E, et al. Dental enumeration and multiple treatment detection on panoramic X-rays using deep learning. *Sci Rep.* 2021;11:12342.
 25. Akpoki V, Martin T, Oluwadare O. EnsembleSplice: ensemble deep learning model for splice site prediction. *BMC Bioinform.* 2022;23:413.
 26. Ensemble Learning Methods for Deep Learning Neural Networks. Machine Learning Mastery. Accessed May 23, 2023. <https://machinelearningmastery.com/ensemble-methods-for-deep-learning-neural-networks/>.
 27. Younis EMG, Zaki SM, Kanjo E, Houssein EH. Evaluating ensemble learning methods for multi-modal emotion recognition using sensor data fusion. *Sensors (Basel).* 2022;22:5611.
 28. Rethinking ImageNet pre-training, 2019. In: He K, Girshick R, Dollar P, eds. *IEEE/CVF International Conference on Computer Vision (ICCV)*, Seoul, South Korea; 2019.
 29. Setzer FC, Shi KJ, Zhang Z, et al. Artificial Intelligence for the computer-aided detection of periapical lesions in cone-beam computed tomographic images. *J Endod.* 2020;46:987-993.
 30. Zheng Z, Yan H, Setzer FC, et al. Anatomically constrained deep learning for automating dental CBCT segmentation and lesion detection. *IEEE Trans Autom Sci Eng.* 2020;18:603-614.
 31. Fatima A, Shafi I, Afzal H, et al. Deep Learning-based multiclass instance segmentation for dental lesion detection. *Healthcare (Basel).* 2023;11:347.
 32. Zhang J, Whebell S, Gallifant J, et al. An interactive dashboard to track themes, development maturity, and global equity in clinical Artificial Intelligence research. *Lancet Digit Health.* 2022;4:e212-e213.
 33. Schwendicke F, Mertens S, Cantu AG, Chaurasia A, Meyer-Lueckel H, Krois J. Cost-effectiveness of AI for caries detection: randomized trial. *J Dent.* 2022;119:104080.